

Muhammad Rehman

AI ENGINEER · MACHINE LEARNING

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Summary

AI Engineer with experience in designing and deploying AI agents, RAG systems, and intelligent applications using advanced LLM frameworks. Skilled in deep learning (Transformers, CNNs, NLP), data preprocessing, and building scalable production-ready solutions with Python, FastAPI, and Docker. Proficient in MLOps practices, MLflow, CI/CD, and AWS, with a passion for developing multimodal AI systems that solve real-world problems.

Tools & Technologies

• Languages & Libraries:

Python – SQL – Langchain – LangGraph – Tensorflow – Scikit-learn – NumPy – Pandas – Matplotlib – FastAPI – Pydantic – Hugging Face

• AI & Machine Learning:

Agents – RAG – LLMs – Fine-Tuning – Transformer – NLP – ANNs – Supervised & Unsupervised Learning – Hyperparameter Tuning – Model Evaluation – EDA – Feature Engineering – MLflow – Docker – CI/CD – EC2,S3

• Tools & Platforms:

VSCode – Jupyter Notebook – Git – GitHub – MySQL – Ollama

Experience

ARCH TECHNOLOGIES || AI Engineer Intern

Remote

- **AI Medical Assistant:** Developed a **multimodal AI chatbot** capable of handling medical-related conversations through text, voice, and image inputs, using LLM-based reasoning and vision capabilities to provide educational responses based on user queries.
 - ◊ **Implementation:** Integrated **Groq-powered** LLM inference, **speech-to-text** for patient voice input, **image analysis** for medical images, and **text-to-speech** to generate AI doctor-style voice responses. Built the interactive application using Gradio with modular Python components for voice, vision, and conversational processing..
 - ◊ **Outcome:** Built an end-to-end **multimodal AI** application combining **LLMs**, computer vision, speech recognition, and text-to-speech into a unified conversational interface, demonstrating practical experience in **Generative AI** and AI application development. .

OTHER PROJECTS

- **Text Summarization App |GitHub**
 - ◊ Developed a **T5-Base** Transformer for automated **text summarization** on large datasets.
 - ◊ Built an **MLOps** workflow covering data ingestion,**transformation**,training,and evaluation, ensuring reproducibility.
 - ◊ Containerized and deployed the **FastAPI** using Docker and integrated with **AWS EC2** for scalable cloud deployment.
 - Tools :** Python, **Transformers**, PyTorch, MLflow ,**Hugging Face**, Pandas, **FastAPI**, Docker,EC2, Git, html, css.
- **Laptop-Prediction-App |GitHub**
 - ◊ Developed a laptop price prediction model using regression based on RAM, processor, storage, and brand.
 - ◊ Performe data preprocessing, feature engineering, and exploratory data analysis to improve prediction accuracy.
 - Tools :** Python, **Scikit-learn**, Pandas, **Docker**,Streamlit,**FastAPI**.

Certifications

UDEMY: MACHINE LEARNING WITH PYTHON

[View Certificate](#)

UDEMY: SQL FOR DATA ANALYSIS

[View Certificate](#)

Academics

Virtual University of Pakistan

B.S. IN COMPUTER SCIENCE

Lahore, Pakistan